

Fin / Digifin

The **tail fin** is the Slocum's steering assembly – and on modern gliders it is a self-contained, self-calibrating **digital fin ("digifin")**. The fin houses far more than the rudder: it integrates the vehicle's **antennas** (Argos 401 MHz, Freewave 900 MHz, and the combined GPS 1575 MHz / Iridium 1626 MHz quad-helix), the **recovery strobe**, the **rudder and its motor**, and a **leak detect** circuit. The whole assembly is rugged enough to be used as a handle to manipulate the glider.



Source

Paraphrased from the *Slocum G3 Glider Operators Manual* (Rev. 1, "Fin" / "Tail Fin" and "Steering Subsystem"), the UG2 community Slack, and the Teledyne Webb Research user forum. This is a condensed field reference – always defer to the official Teledyne documentation and the **4095-FCP Functional Checkout Procedure** for your specific glider.

What's in the fin

The G3 fin is built around a **titanium boom**, a **motor housing**, an antenna feed-through, and a plate bolted to a **PEEK mast** with a barrel-nut fastener system. The rudder rides on a **magnetic coupling** (an external magnet drives an internal one – there is no shaft seal at the rudder), and the strobe is integrated into the leading edge of the mast in the orange Noryl housing shared with the GPS/Iridium antenna.

Four independent subsystems live in the fin and should each be tested (per the FCP):